

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF
ECONOMICS Hypothesis Testing I

Week 10 · Averages and Proportions · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 Academic
Year



Second question

What did week 9 answer

“How much is μ ?”

Answer: [4.4600; 4.8100] — the estimate and its uncertainty together.

What does week 10 ask

“Is μ equal to such and such a number?”

This is a different question; now the answer is not a number, but a decision. And every decision can be wrong.

Hypothesis testing is not a new idea. It is a different way of looking at the confidence interval. We will prove this in the first slide.

A common mistake However, the μ -value is the most misunderstood concept in statistics. Half of this week is devoted to it.



Learning Objectives for Week 10

By the end of this week, you will:

1. Formulate the Null Hypothesis and Alternative Hypothesis before viewing the data.
2. Calculate the p -value using a simulation — without a formula — and state its definition.
3. Perform a t -test for the mean and a z -test for the proportion.
4. Calculate Type I Error, Type II Error, and Power.
5. **Reject the incorrect interpretations of the p -value numerically.**
6. You will recognize and measure p -hacking.
7. You will distinguish statistical significance from practical significance.

What exactly does this mean — and what does it not mean?

CLO7 — Formulate hypotheses, test them, and interpret the p -value correctly. Bloom's: Analysis / Evaluation.

The method remains the same: we know the original set, so the error of each method is not stated, but counted. This week we will even simulate a world where H_0 is true — and see what happens there.

**Interval and test are the same
thing**

The Promised Bridge — Proof

Sample from week 9: Uzbekistan wheat yield 2009-2018, $\bar{x} = 10$, $s = 4.6350$, $n = 0.2446$, $SE = 0.0774$. 95% interval: $[4.4600; 4.8100]$.

Is $\mu \leq 0$ in the interval?

4.30 4.331 0.0019 rejected no 4.40 3.038 0.0141
rejected no 4.460019 2.262157 0.050000 boundary
boundary 4.50 1.745 0.1149 not rejected yes 4.635
0.000 1.0000 not rejected yes 4.809981 -2.262157
0.050000 boundary boundary 4.90 -3.426 0.0076
rejected no

Theorem Exactly, not approximately. At the unrounded end of the interval, μ is exactly 0.050000. At the rounded 4.4600, $\mu = 0.049980$ — formally rejected.

Confidence interval = the set of all μ that cannot be rejected. Nothing else.

Formulating hypotheses

Two hypotheses — not equal

H_0 — null hypothesis: “nothing”, “the difference is zero”, “ $\mu = \mu_0$ ”. Always with equality.

H_a — alternative: what the researcher wants to show. $\neq$, $<$ or $>$.

Common mistake

The asymmetry is intentional. As in a trial: the accused is presumed innocent, and only strong evidence will change that.

Therefore, “we could not disprove H_0 ” never means “ H_0 is correct” — just as “not proven guilty” does not mean “not guilty”.

Three hard rules

1. Hypotheses are about parameters (μ , σ), not about statistics. “ $\bar{x} - \bar{y} = 4.5$ ” is meaningless.

2. The direction (one-sided or two-sided) is chosen before seeing the data. Then the choice is μ -hacking.

3. μ is also predetermined.

Audience question

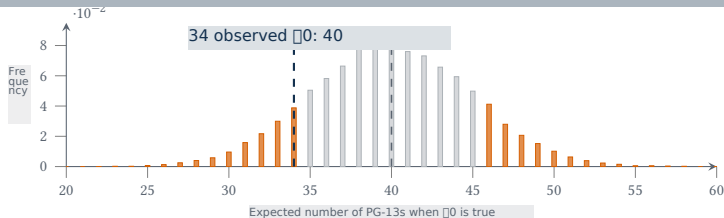
Question. We want to test the claim that “the yield of the new variety is higher than the old one.” What are H_0 and H_a ?

$H_0: \mu = \mu_{\text{eski}}$; $H_a: \mu > \mu_{\text{eski}}$. The claim is in the alternative, not in the null.

[-value

$\hat{p}$ -value — first without formula

Question: Are 40% of movies PG-13? $H_0: p = 0.40$, $H_A: p \neq 0.40$. Sample: $n = 100$, 34 observed, $\hat{p} = 0.34$.



Protocol — complete

1. Assume p_0 is true: each movie is PG-13 with probability 0.40.
2. "Create" 100 movies using this rule.
3. Repeat this 20,000 times $\rightarrow$ null distribution.
4. Count the results as observed or more extreme.

Deviation $|34 - 40| = 6$. Deviation greater than or equal to this is 5,232 times out of 20,000: $\hat{p} = 5,232/20,000 = 0.2616$. The exact binomial value is 0.2614.



Audience Question

No formula used — just the question "What if p_0 were true" 20,000 times.

Definition of the p -value — and transition to the formula

The probability of getting the observed result or a more extreme result, given that the H_0 -value is true: $p = P(\text{the same or more extreme data} | H_0)$.

The same question, with four methods:

Method

Simulation (20,000)
0.2616 Exact binomial
0.2614 p -test 0.2207 p -test
+ correction 0.2616

The p -test without correction is 15.6% lower than the exact value; with correction, the difference is 0.1%.

p -test for proportion

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

p_0 in denominator, NOT $\hat{p}$ — we are calculating the world-ode where p_0 is correct. $\hat{p} = 0.34 - 0.40$

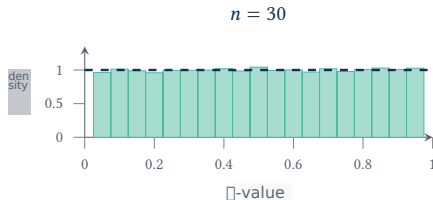
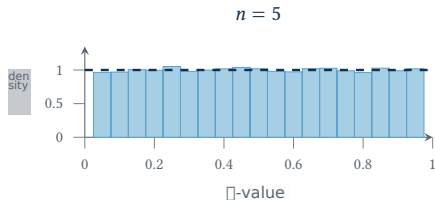
$$= -1.2247$$

0.34 - 0.40

Common error

Most common technical error: calculating SE from $\hat{p}$ in test. $\hat{p}$ in interval, p_0 in test.

Key to the whole topic: θ is uniformly distributed under θ_0



40,000 θ -tests with θ_0 being true. θ does not favor any value.

Theorem Measured: The proportion of $\theta < 0.05$ is 0.0482 and 0.0481; the proportion of $\theta < 0.01$ is 0.0100 and 0.0093.

A common error

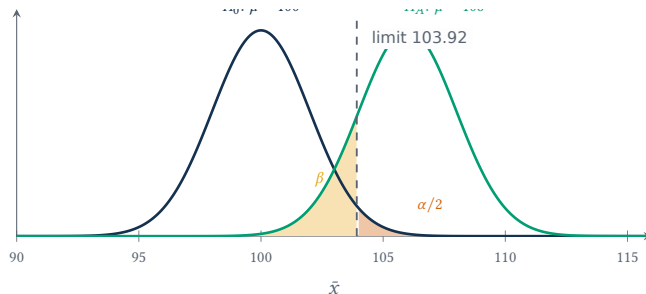
Consequence: Even if θ_0 is true, one in every 20 tests will give $\theta < 0.05$.

Audience Question

This one fact explains everything else this week: why θ -hacking works and why one “significant” result proves nothing.

Errors and power

Two types of errors — and they are interchangeable



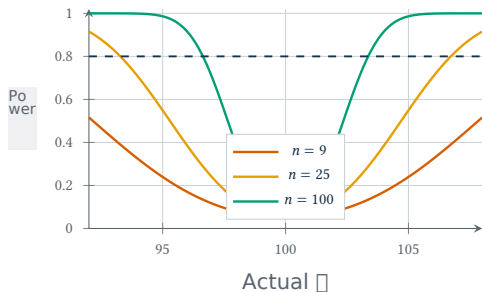
α 0 correct β 0 incorrect

rejected Type I (α) correct $(1 - \alpha)$
not rejected correct Type II (β)

Common error

If you make α smaller, β increases ($\alpha = 36$, $\beta = 106$): $\alpha = 0.10$
 $\rightarrow$ power 0.912; $\alpha = 0.01 \rightarrow 0.664$; $\alpha = 0.001 \rightarrow 0.386$. Only β
improves both together.

Power: What can we see?



$\mu_0 : \mu = 100$, $\mu = 12$, $\alpha = 0.05$. Dotted line — typical Target 0.80.

Required n for 80% power

$$n \geq \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{d} \right)^2, \quad d = \frac{\delta}{\sigma}$$

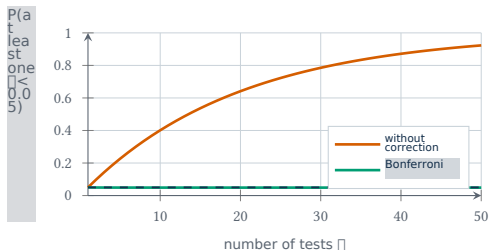
δ	d	n
2	0,167	283
3	0,250	126
4	0,333	71
6	0,500	32
12	1,000	8

Audience Question

When the effect is halved, it is n quadrupled — again the $\sqrt{n}$ law (week 7). Power is calculated BEFORE the study.

[-hacking

The first form of α -hacking: multiple testing



H_0 is true everywhere: finding a “discovery” from pure noise probability.

A common mistake

m	$1 - 0.95^m$
1	0,050
5	0,226
14	0,512
20	0,642
50	0,923

In 20 tests, the “discovery” is more than likely. Simulation 0.6411 (theoretical 0.6415).



Bonferroni theorem: α/π for each test. $\pi = 20$: $\alpha = 0.0025$; the total error remains at 0.0488. More details — in week 11.

Second and third forms: stopping and searching

Violating the stopping rule

The researcher repeats the test after each new observation and stops when $p < 0.05$. $p=0$ is always true.

The mean is a true type I error	
10 → 30	0,172
10 → 100	0,282
10 → 200	0,348

The declared 0.05 is actually 3-7 times larger.

Searching by subgroups

General test + halving on 4 insignificant signs = 9 tests, all on the same data.
Simulation (pure noise): 0.2624 — 5.2 times the declared 0.05.

Theorem 1 $-0.959 = 0.370$ is NOT an error. These 9 tests are not independent, and the dependence makes the formula highly estimator. In the previous slide, the tests were independent — there the formula was exactly the same.

Audience Question

The only defense is preregistration: the hypothesis, α , β , direction, and analysis plan are written before the data collection.

□ **what is it?**

— NOT “the probability that H_0 is true”

A common mistake

$\beta = P(\text{data} | H_0)$, but not $P(H_0 | \text{data})$.

This is exactly the prosecutor’s fallacy from week 3. Reading the conditional probability backwards.

The Sellke-Bayarri-Berger (2001) bound gives the most favorable case: minimum Bayes factor = $-\beta \ln \beta$.

β min Bayes factor $P(H_0 | \text{data})$

0,05	0,4072	$\geq 0,289$
0,01	0,1252	$\geq 0,111$
0,001	0,0188	$\geq 0,018$

(The initial probability is taken to be 0.5.)

Theorem H_0 is at least 29% more likely than not to be true when $\beta = 0.05$.

And this is a reasonable estimate. In practice, it is usually much higher.

A common mistake

Initial confidence is important. If the hypothesis is initially doubtful ($P(H_0) = 0.10$):

$\beta P(H_0 \text{data})$	
0,05	0,786
0,01	0,530
0,001	0,145

An implausible claim cannot be proven with $\beta = 0.05$.

Statistical significance $\neq$ practical significance

True population: $\mu = 35.805$, $\mu = 37.405$ known
— hence $\mu_0 : \mu = 34$; true difference 1.805,
Cohen $d = 0.048$ (nonsignificant).

μ μ_0 conclusion

30	0.264	0.792	insignificant	100
0.483	0.629		insignificant	400
0.965	0.334		insignificant	1000
1.526	0.127		insignificant	2000
2.158	0.031		SIGNIFICANT	

The effect has not changed — only n has increased. At sufficiently large n , any difference from zero becomes “significant”.

There is also a reverse error HYPOTHESIS: the real difference is \$15 million ($d = 0.40$ — large), but $n = 10$. $\mu = 1.268$, $\mu_0 = 0.205$, power 0.245. (If d is unknown, n_9 is used, $n = 0.237$ — always state which convention.)

“Not significant” does not mean “no difference,” it means “we couldn’t see it.”

Theorem Rule. Never give p alone.
Always give all three together:

effect size + confidence interval + p

Misconception Why it's wrong

" $\alpha = 0.05$ — probability that H_0 is true." $\alpha = P(\text{data} | H_0)$. The inverse is at least 0.289. Prosecutor's fallacy (week 3). " $\alpha > 0.05$ — so H_0 is true." "Insufficient evidence" $\neq$ "evidence has been proven to be lacking." Even with $\alpha = 0.40$, the power at $\alpha = 10$ is only 0.245. " α small — effect large." α does not measure effect size. At $\alpha = 2000$, $\alpha = 0.048$ also gives $\alpha = 0.031$. " $\alpha = 0.049$ and $\alpha = 0.051$ are radically different." 0.05 is a convention, not a law of nature. α is uniformly distributed below H_0 ; there are no jumps around the boundary. "You can choose the best of several tests." The probability of "discovery" from pure noise in 20 tests is 0.64. The direction and α are recorded in advance. "I will repeat the test until I have enough data." This increases the Type I error from 0.05 to 0.35.

"We have proven H_0 ." The test never proves H_0 . It only rejects or fails it.

**Summar
y**

Must remember

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

$$z = \sqrt{n}(\hat{\mu} - \mu_0) / \hat{\sigma} \quad (\mu_0, \hat{\sigma} \text{ known})$$

$$n \geq \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{d} \right)^2, \quad d = \frac{\delta}{\sigma}$$

Decision: $\hat{\mu} < \hat{\mu}_0 \Rightarrow$ rejected. Equally strong: Is $\hat{\mu}_0$ outside the interval?

In three sentences

1. The test is the other side of the confidence interval. There is no new math, there is a new decision.

2. $\hat{\mu}$ — the answer to the question “How unusual is the data if μ_0 is true?” Nothing else.

3. $\hat{\mu}$ is uniformly distributed under μ_0 — so a single small $\hat{\mu}$ proves nothing, and so it is easy to find by searching.

Next Step

This week we worked with one sample. In week 11, there will be two: two-group difference, paired observations, and chi-square for categorical data. The problem of multiple comparisons is also fully addressed there.

References

Main reading

- Cetinkaya-Rundel M., Hardin J. Introduction to Modern Statistics (IMS, 2nd ed.), Chapters 11 and 17–19 — simulation-based testing first, formula later. Open source: openintro.org/book/ims
- Newbold P., Carlson W., Thorne B. Statistics for Business and Economics (9th ed.), Chapter 9.

This week's required reading — both short

- Wasserstein R., Lazar N. (2016). The ASA statement on p -values: context, process, and purpose. The American Statistician 70(2), 129–133. — six principles, 2 pages.
- Greenland S. et al. (2016). Statistical tests, p values, confidence intervals, and power: a guide to misinterpretations. European Journal of Epidemiology 31, 337–350. — 25 misinterpretations of the p -value, one paragraph each.

Supplement

- Simmons J., Nelson L., Simonsohn U. (2011). False-positive psychology. Psychological Science 22(11), 1359–1366. — An article showing how p -hacking works.
- Sellke T., Bayarri M., Berger J. (2001). Calibration of p values for testing precise null hypotheses. The American Statistician 55(1), 62–71.
- Ioannidis J. (2005). Why most published research findings are false. PLoS Medicine 2(8), e124.